so formed are called ionic compounds. A compound is said to be ionic when the charge of the cation and anion are completely separated. But in 1923, Kazimierz Fajans found, through his X-Ray crystallographic studies, that some of the ionic compounds show covalent character. Based on this, he formulated a set of rules to predict whether a chemical bond is ionic or covalent. Fajan's rules are formulated by considering the charge of the cation and the relative size of the cation and anion.

- When the size of the cation is small and that of anion is large, the bond is of more covalent character
- Greater the charge of the cation, greater will be the covalent character

| Ionic | Covalent |
|---|---|
| Low positive charge | High positive charge |
| Large cation | Small cation |
| Small anion | Large anion |

For example, in sodium chloride, low positive charge (+1), a fairly large cation and relatively small anion make the charges to separate completely. So it is ionic. In aluminium triiodide, higher is the positive charge (+3), larger is the anion and thus no complete charge separation. So is covalent. The following picture depicts the relative charge separation of ionic compounds:

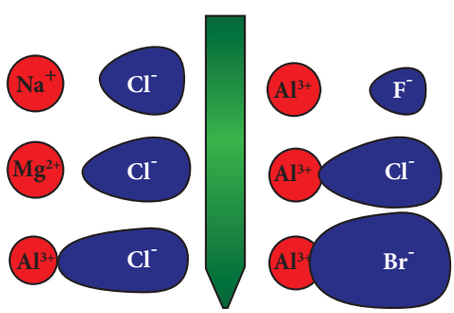



**Figure. 13.14** Relatice charge separation of ionic compounds

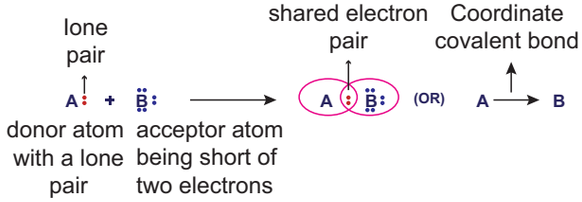
## 13.3.3 Coordinate covalent bond

In the formation of normal covalent bond each of the two bonded atoms contribute one electron to form the bond. However, in some compounds, the formation of a covalent bond between two atoms takes place by the sharing of two electrons, both of which comes from only one of the combining atoms. This bond is called ***Coordinate covalent bond or Dative bond***.

Mostly the lone pair of electrons from an atom in a molecule may be involved in the dative bonding. The atom which provides the electron pair is called **donor atom** while the other atom which accepts the electron pair is called **acceptor atom**. The coordinate covalent bond is represented by an arrow (→) which points from the donor to the acceptor atom.

**Formation of coordinate covalent bond**

Let us consider two atoms A and B. Let atom A has an unshared lone pair of electrons and atom B is in short of two electrons than the octet in its valence shell. Now atom A donates its lone pair while atom B accepts it. Thus the lone pair of electrons originally belonged to atom A are now shared by both the atoms and the bond formed by this mutual sharing is called Coordinate covalent bond. (A→B)




Examples ($NH^{4+}$, $NH_3 \rightarrow BF_3$)

**Illustration 1 – Formation of coordinate covalent bond between $NH_3 \rightarrow BF_3$ molecules**

In some cases, the donated pair of electrons comes from a molecule as a whole which is already formed to another acceptor



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